

14. Test Closure Report

14.1 Introduction

Test Closure is the final phase of the Software Testing Life Cycle (STLC). It is performed after all planned testing activities have been completed. The purpose of test closure is to evaluate testing outcomes, document results, assess quality objectives, identify lessons learned, and formally conclude the testing process.

This document summarizes all testing activities performed for the Parabank Banking Application Automation Project, including manual testing, automation testing, defect management, and CI/CD execution.

14.2 Project Overview

Item	Details
Project Name	Parabank Automation Framework
Application	Parabank Banking Application
Testing Type	Functional Testing
Automation Tool	Selenium WebDriver
Framework	TestNG
CI/CD Tool	GitHub Actions
Version Control	Git & GitHub

14.3 Testing Scope Covered

The following modules were tested:

Login Module

- Valid Login
- Invalid Login
- Blank Credentials
- Error Message Validation

Account Overview Module

- Account Balance Verification
- Account Summary Validation
- Account Details Verification

Open New Account Module

- Savings Account Creation
- Checking Account Creation
- Confirmation Validation

Transfer Funds Module

- Valid Transfer
- Invalid Transfer
- Transfer Confirmation

Bill Pay Module

- Bill Payment
- Payee Validation
- Transaction History Update

Registration Module

- New User Registration
- Mandatory Field Validation
- Duplicate User Validation

Find Transactions Module

- Search by ID
- Search by Date
- Search by Amount

Request Loan Module

- Loan Approval
- Loan Rejection
- Confirmation Validation

Contact Information Module

- Address Update
- Phone Update
- ZIP Code Update

14.4 Test Execution Summary

Metric	Count
Total Test Scenarios	50
Total Test Cases	200
Passed Test Cases	191
Failed Test Cases	9
Blocked Test Cases	0
Not Executed	0

14.5 Defect Summary

Severity	Count
Critical	0
High	2
Medium	4
Low	3
Total Defects	9

14.6 Automation Summary

Automation framework successfully implemented using:

- Selenium WebDriver
- Java
- TestNG
- Maven
- Page Object Model

Automated Modules:

- Login
 - Account Overview
 - Transfer Funds
 - Bill Pay
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14.7 CI/CD Execution Summary

CI/CD pipeline successfully implemented using GitHub Actions.

Pipeline Activities:

- Source code checkout
- Java setup
- Maven build
- Selenium execution
- Report generation

Execution Status:

✓ Successful

14.8 Deliverables Completed

The following deliverables were completed successfully:

- Project Introduction
- Requirement Analysis
- Test Plan
- Test Scenarios
- Test Cases
- RTM
- Environment Setup
- Automation Framework
- CI/CD Pipeline
- Execution Report
- Defect Report
- Test Closure Report

14.9 Exit Criteria Verification

Exit Criteria	Status
All Test Cases Executed	Pass
Automation Scripts Executed	Pass
Reports Generated	Pass
Defects Logged	Pass
CI/CD Pipeline Validated	Pass
Project Deliverables Completed	Pass

14.10 Challenges Faced

During project execution, the following challenges were encountered:

- Chrome browser compatibility issues.
- GitHub authentication setup.
- Git configuration errors.
- GitHub Actions workflow configuration.
- Headless browser execution in CI environment.
- Dependency management issues.

All identified issues were successfully resolved.

14.11 Lessons Learned

Key learnings from this project:

- Importance of requirement analysis.
 - Benefits of Page Object Model.
 - Effective use of data-driven testing.
 - Git and GitHub workflow management.
 - CI/CD implementation using GitHub Actions.
 - Automated test reporting.
 - Defect tracking and management.
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14.12 Project Outcome

The project successfully achieved its objectives.

Achievements

- Automation framework developed.
 - Banking modules tested.
 - Test scenarios and test cases prepared.
 - Defects identified and documented.
 - CI/CD pipeline implemented.
 - Automated reports generated.
 - Continuous testing enabled.
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14.13 Recommendations

Future improvements may include:

- API Automation Testing
- Database Validation

- Cross Browser Testing
 - Docker Integration
 - Jenkins Integration
 - Performance Testing
 - Security Testing
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14.14 Conclusion

The testing activities for the Parabank Banking Application were completed successfully. All planned test scenarios, test cases, automation scripts, and CI/CD activities were executed according to the project objectives. The automation framework improved testing efficiency, reduced manual effort, and enabled continuous testing through GitHub Actions. The project demonstrated the successful implementation of an end-to-end automation and CI/CD solution aligned with industry-standard software testing practices.